

Civilization and The State of the World: Year 2027

1. We Live in the Anthropocene

- * Humans are now the main drivers of biogeophysical change on Earth, for nearly all biogeophysical processes and at nearly all spatial scales.

- * The Anthropocene is a proposed new geological epoch based on the observation that the magnitude, variety and longevity of human-induced changes on essential planetary processes have become so profound that we are no longer in the Holocene.

- * Human activity is now the driving force in determining the trajectory of the entire Earth System.

- * Anthropocene begins in the geologic record in either 1610 or in 1964:
 - 1610 (Orbis spike) – CO₂ drop from population crash in the Americas; highlights social concerns, unequal power relationships between different groups of people, economic growth, the impacts of globalized trade, and our current reliance on fossil fuels. The onward effects of the arrival of Europeans in the Americas also highlights a long-term and large-scale example of human actions unleashing processes that are difficult to predict or manage.
 - 1964 (atomic bomb spike) – an elite-driven technological development that threatens planet-wide destruction; highlights the Great Acceleration since the 1950s which is marked by a major expansion in human population, large changes in natural processes, and the development of novel materials from minerals to plastics to persistent organic pollutants and inorganic compounds, including these the global fallout from nuclear bomb tests.

Steffen et al., 2018

Lewis and Maslin, 2015

2. State of the Earth: Year 2027

- * Rapidly moving to Hot House Earth, freshwater depletion, melted glaciers, Sixth Mass Extinction,
 - (a) The Atmosphere
 - (b) The Hydrosphere
 - (c) The Cryosphere
 - (d) The Biosphere

(e) The Lithosphere

- * Triggered seismicity from mining and extraction

- * Subsidence, sinking, and erosion effects from the built environment

McGarr et al., 2002

Meijer et al., 2018

3. State of the Humans: Year 2027

(a) Current Societal Organizing Principles

- i. Money

 - A. Capital and Wealth

 - B. Debt-Based, Neo-Liberal Economics

- ii. Status and Power

- iii. Religion

- iv. Government and Empire

(b) Current Technology

(c) Current Population

4. State of Civilization: Year 2027

(a) Globalized

(b) Weaponized

(c) Distracted

(d) Fragile

REFERENCES

- Lewis, S. L., & Maslin, M. A. (2015). Defining the anthropocene. *Nature*, 519(7542), 171–180 (cit. on p. 1).
- McGarr, A., Simpson, D., & Seeber, L. (2002). Case histories of induced and triggered seismicity. In *International geophysics* (pp. 647–661, Vol. 81). Elsevier. (Cit. on p. 2).
- Meijer, J. R., Huijbregts, M. A. J., Schotten, K. C. G. J., & Schipper, A. M. (2018). Global patterns of current and future road infrastructure. *Environmental Research Letters*, 13(6), 064006 (cit. on p. 2).
- Steffen, W., Rockström, J., Richardson, K., Lenton, T. M., Folke, C., Liverman, D., Summerhayes, C. P., Barnosky, A. D., Cornell, S. E., Crucifix, M., et al. (2018). Trajectories of the earth system in the anthropocene. *Proceedings of the national academy of sciences*, 115(33), 8252–8259 (cit. on p. 1).